General
MODEL EXAM Batch 2025 CSE...
Finding Time Complexity of ...
Problem 1: Finding Complexit...
Problem 2: Finding Complexit...
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Problem 5: Finding Complexit...
Greedy Algorithms
1-G-Coin Problem
2-G-Cookies Problem
3-G-Burger Problem
4-G-Array Sum max problem
5-G-Product of Array element...
Divide and Conquer
1-Number of Zeros in a Given ...

Problem 5: Finding Complexity using counter method

Started on	Monday, 24 August 2026, 1:48 PM
State	Finished
Completed on	Monday, 24 August 2026, 2:00 PM
Time taken	11 mins 42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 Correct Mark 1.00 out of 1.00 Flag question

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
   int rev = 0, remainder;
   while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n/= 10;

    }
print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n
Output:
Print the value of the counter variable

Answer:

```c

#include <stdio.h>

void reverse(int n)
{
    int rev = 0, remainder;
    int count = 0;

    while (n != 0)
    {
        count++;  // while condition

        remainder = n % 10;
        count++;

        rev = rev * 10 + remainder;
        count++;

        n /= 10;
        count++;
    }

    count++;  // final failed while condition

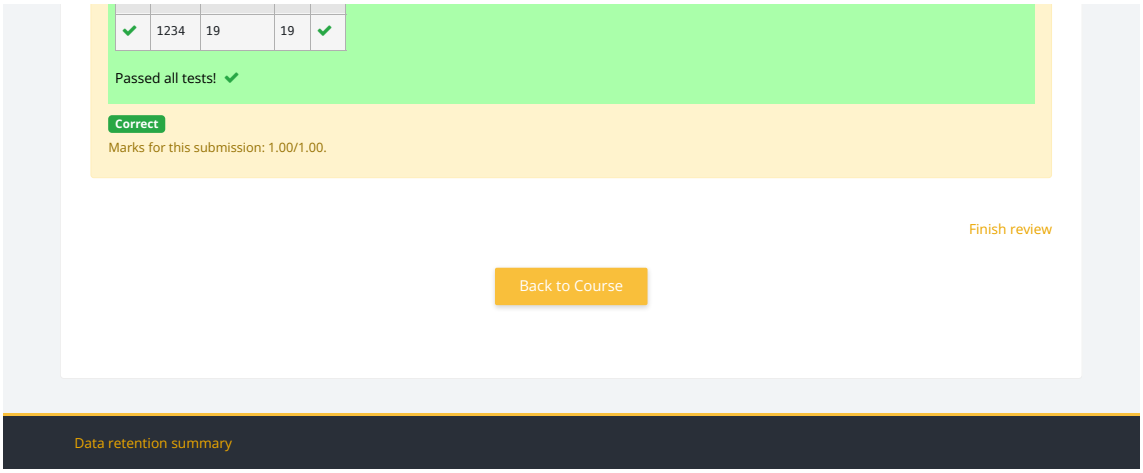
    printf("%d", count+2);//it is for the two initalisations
}

int main()
{
    int n;
    scanf("%d", &n);

    reverse(n);

    return 0;
}
```

	Input	Expected	Got	
✔	12	11	11	✔



✔	1234	19	19	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Finish review

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Data retention summary